

Nicholas Karantakis

nicholaskarantakis@gmail.com | www.linkedin.com/in/nicholas-karantakis | nicholaskarantakis.com

Education

University of Toronto | Cumulative GPA 3.91/4.0

Sept. 2023 – Apr. 2028

Bachelor of Applied Science and Engineering (B.A.Sc.) in Mechanical Engineering + PEY Co-op

Toronto ON

Intended Minors: Robotics and Engineering Business

Experience

Hardware Vision Engineering Intern

Sept. 2025 – Aug. 2026

Martinrea International Inc. - Advanced Manufacturing Technology Group

Vaughan ON

- Designed and installed computer vision system for various applications in automotive manufacturing lines around the world.
- Documented the process flow and scope for each plant project by regularly conducting meetings with various production and quality teams.
- Aided in each step of system integration process including building vision panels, model training and validation, hardware installation, camera and lighting configuration, UI development, configuring network/IP settings, system troubleshooting and more.

Lead-Free Piezoelectric Nanogenerator Research

May. 2025 – Aug. 2025

Mechatronics and Microsystems Design Laboratory

Toronto ON

- Conducted research on lead-free piezoelectric nanogenerators (PENG) by doping and synthesizing Zinc Oxide nanowires with various elements to enhance piezoelectric efficiency for sustainable energy harvesting.
- Designed and fabricated custom testing apparatus using SolidWorks and in-house machining tools; performed soldering and assembly to support voltage output measurements of doped Zinc Oxide nanowires.
- Developed and optimized experimental setups for electrical characterization of piezoelectric materials, resolving technical issues and refining testing methodologies to improve data accuracy and reproducibility.

Airframe Member

Jan. 2024 – May. 2025

University of Toronto Aerospace Team - UAS AEAC Drone Team

Toronto ON

- Facilitated the design, building, and testing of airframe components for the AEAC competition.
- Developed and tested engineered point of failure for motor mounts to protect propellers in case of a crash.
- Created design for drone shell in SolidWorks using surface modeling and developed mold for fiberglass layups

Lung Airway and Vasculature Research

May. 2024 – Aug. 2024

Latner Thoracic Research Laboratories and Bazylak Group

Toronto ON

- Optimized MATLAB vessel network extraction pipeline using MATLAB Profiler tools and memory allocation techniques to achieve 47% reduction in process time for extracting the geometric characteristics of lung vasculature and airways from image stacks.
- Developed imaging techniques for mouse lung airways based on casting procedures from literature using MICROFIL and silicone rubber casting material to produce micro-computed tomography images with over 1000 branch segments up to ~30 generations.
- Constructed simplified airway geometries for Computational Fluid Dynamics analysis from image data using Python and Blender.

Production Support

July. 2022 – Aug. 2022

Mitsubishi Heavy Industries Canadian Aerospace

Mississauga ON

- Contributed to Bombardier Global Express/5000 wings and center fuselage production.
- Supported various tasks, including paint touchups, cleaning, sandblasting, and preparing wings for leak tests.
- Enhanced soft skills such as adaptability and communication by assuming diverse roles at all production stages including the spar shop, final inspection, and preparation for shipping.

Skills

Programming: MATLAB, Python, JavaScript, Arduino C++, HTML/CSS

Frameworks: React, Node.js, Flask

Libraries: Git, Pandas, NumPy, Matplotlib, Network X, Media Pipe, Open CV, PyQt5, Mapbox, YOLO, BPY

CAD / 3D Modelling: SolidWorks CSWA, Blender, 3D Slicer

Machining: Lathe, Mill, Drill Press, Laser Cutting

Office Tools: Excel, Word, PowerPoint

Personal: Public Speaking, Adaptability, Problem-Solving, Teamwork